UNIVERSITY OF BIRMINGHAM

School of Physics and Astronomy

DEGREE OF M.Sci. WITH HONOURS

FOURTH-YEAR FINAL EXAMINATION

03 17307

LM Superconductivity

SEMESTER 1 EXAMINATIONS 2021/22

Time Allowed: 1 hour 30 minutes

Answer ***two*** questions. If you answer more than two questions, credit will only be given for the best two answers.

The approximate allocation of marks to each part of a question is shown in brackets **[ ]**.

All symbols have their usual meanings.

A list of useful formulae is provided at the end of the paper.

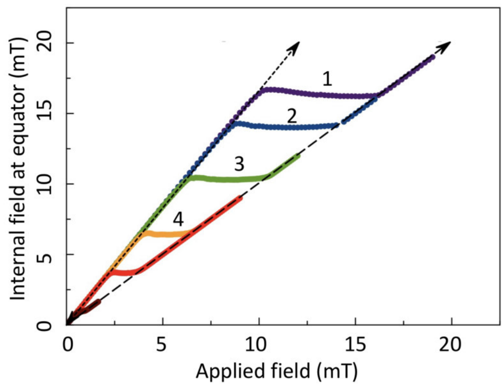
Calculators may be used in this examination but must not be used to store text. Calculators with the ability to store text should have their memories deleted prior to the start of the examination.

Physical constants and units that may be required will be found at the end of this question paper.

Answer **two** *questions. If you answer more than two questions, credit will only be given for the best two answers.*

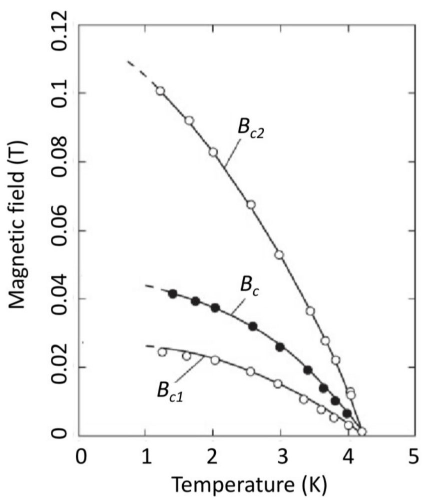
1. Superconductors can be classified into either type-I or type-II depending on the relative scale of the penetration depth λ and the coherence length ξ.

   (a) The figure below shows the internal magnetic field on the surface of an ellipsoidal-shaped type-I superconductor at the equator as a function of the applied field along the long axis of the ellipsoid. Different curves represent the data obtained at different temperatures. The curves marked by numbers from 1 to 4 represent those obtained at 2.0, 2.2, 2.5 and 2.8 K, respectively.



   Interpret the data and briefly explain what is happening physically. Then, estimate the thermodynamic critical field of the above superconductor at zero temperature. **[10]**

   (b) The figure below shows the field-temperature phase diagram of an alloy.



The upper critical field B_{c2} may be defined as the field when the flux through an area set by ξ^2 equals the flux quantum, Φ_0. Express this argument in a mathematical form (*Note: The requested mathematical form could be derived from solving the Ginzburg-Landau equation, but here we wish to arrive at the solution without resorting to that equation*). Then, estimate the values of ξ and λ for this alloy at zero temperature. **[10]**

(c) Consider a vortex in a superconducting alloy within the London theory. Suppose that the core of the vortex, which is essentially normal metallic, has a radius of the order of ξ.

Use $\nabla \times (\nabla \times \mathbf{B}) = -\mathbf{B}/\lambda^2$, to show that the magnetic field outside the core is described by the following Bessel equation,

$$\frac{1}{r}\frac{d}{dr}\left(r\frac{dB_z}{dr}\right) = \frac{B_z}{\lambda^2}.$$

You may use that in polar coordinate,

$$\nabla \times \mathbf{B} = \frac{1}{r}\begin{vmatrix} \hat{r} & r\hat{\phi} & \hat{z} \\ \frac{\partial}{\partial r} & \frac{\partial}{\partial \phi} & \frac{\partial}{\partial z} \\ B_r & rB_\phi & B_z \end{vmatrix}$$

Simplify the above Bessel equation for $\xi < r \ll \lambda$ and then solve to obtain the expression of B_z and the current density $\mathbf{j}$, where $\mu_0\mathbf{j} = \nabla \times \mathbf{B}$. Also simplify the Bessel equation for $r \gtrsim \lambda$ and then solve to obtain an expression for B_z. **[10]**

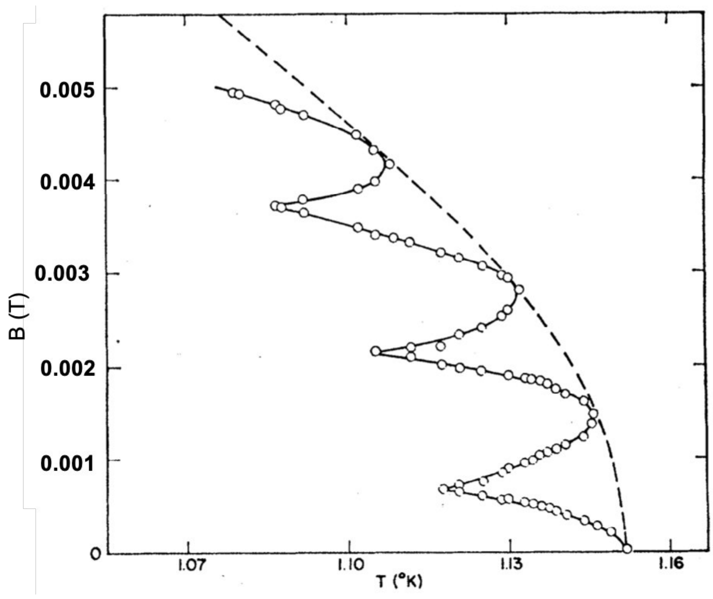
2. Suppose that a thin-walled superconducting cylinder, whose thickness is less than the penetration depth, is placed in a magnetic field $\mathbf{B}$ along the axial direction and its transition temperature T_c is measured as a function of the field. This is called the Little-Parks experiment.

   (a) Use the expression for the supercurrent velocity,
   $$\mathbf{v}_s = \frac{1}{2m_e}\left(\hbar\nabla\phi + 2e\mathbf{A}\right),$$
   to show that
   $$v_s = \frac{\hbar}{2m_eR}\left(n + \frac{\Phi}{\Phi_0}\right),$$
   where R is the radius of the cylinder, n is an integer, Φ is the applied magnetic flux threading the interior of the cylinder and Φ_0 is the flux quantum. **[8]**

   (b) By considering current-carrying solution, $\psi(\mathbf{r}) = |\psi(\mathbf{r})|e^{i\mathbf{k}\cdot\mathbf{r}}$, of the Ginzburg-Landau equation,
   $$-\frac{\hbar^2}{4m_e}\nabla^2\psi + a(T)\psi + b|\psi|^2\psi = 0,$$
   show that the order parameter magnitude, $|\psi|^2$, varies with the supercurrent velocity according to the equation
   $$|\psi|^2 = -\frac{a(T)}{b}\left(1 - \frac{m_e v_s^2}{|a(T)|}\right),$$
   [4]

   (c) Use the above results from part (a) and (b) to show that a superconducting-to-normal transition occurs when
   $$\frac{1}{\xi^2} = \frac{1}{R^2}\left(n + \frac{\Phi}{\Phi_0}\right)^2,$$
   where $\xi = \sqrt{\hbar^2/4m_e|a(T)|}$.
   Use the expression for the temperature dependence of the coherence length
   $$\xi = \xi(0)|t|^{-1/2},$$
   where $t = (T - T_{c0})/T_{c0}$, and T_{c0} is the critical temperature in zero field, to show that
   $$T_{c0} - T_c = T_{c0}\frac{\xi(0)^2}{R^2}\left(n + \frac{\Phi}{\Phi_0}\right)^2.$$
   where T_c is the new critical temperature in the presence of a magnetic field.
   [8]

(d) The figure below shows the data from a Little-Parks experiment using a conventional superconducting metal (dashed line represents the fit to the T_c envelope). The radius of the cylinder is 0.65 μm.



Draw schematic graphs showing resistivity ρ versus field B at 1.16, 1.13 and 1.09 K.

Estimate the zero-temperature coherence length. **[10]**

3. The BCS ground-state wavefunction can be written as

$$|\mathtt{BCS}\rangle = \Pi_{\mathbf{k}} \left(u_k + v_k c^\dagger_{\mathbf{k}\uparrow} c^\dagger_{-\mathbf{k}\downarrow} \right) |0\rangle,$$

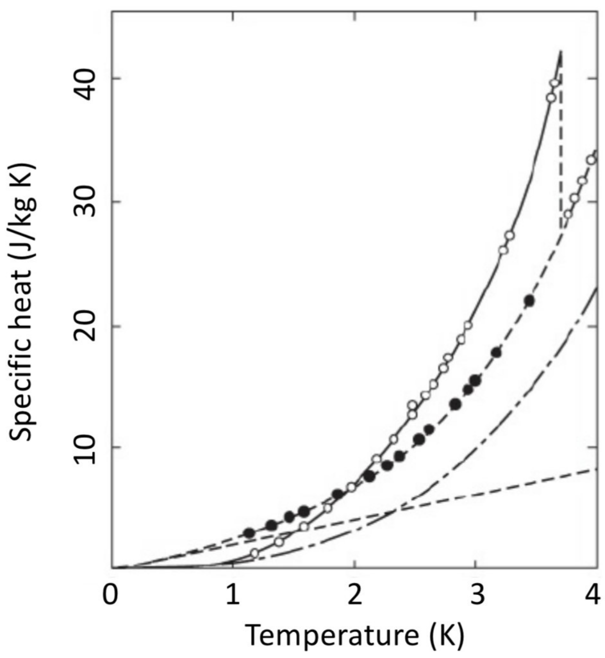
where $|0\rangle$ is the state vector for a vacuum and $c^\dagger_{\mathbf{k}\uparrow}$ is the creation operator of an electron with momentum $\hbar k$ and spin up. The predictions from BCS theory include that i) the electronic specific heat C_e shows a jump at the transition temperature T_c as temperature is lowered, ii) the size of the energy gap Δ at zero temperature is proportional to T_c and iii) the expression for T_c is given by

$$k_{\rm B} T_c = 1.14 \hbar\omega_{\rm D} \exp\left(-\frac{1}{V_0 D(E_{\rm F})} \right).$$

(a) Explain the implications of the above expression for T_c in regard to the superconductivity of different elements. In particular, address the isotope effect. Explain why metals that have high conductivity at room temperature are often not superconducting at low temperature, while low-conductivity metals become superconductors at relatively high temperature, and why 4d and 5d transition metals are often superconductors with relatively high T_c. **[8]**

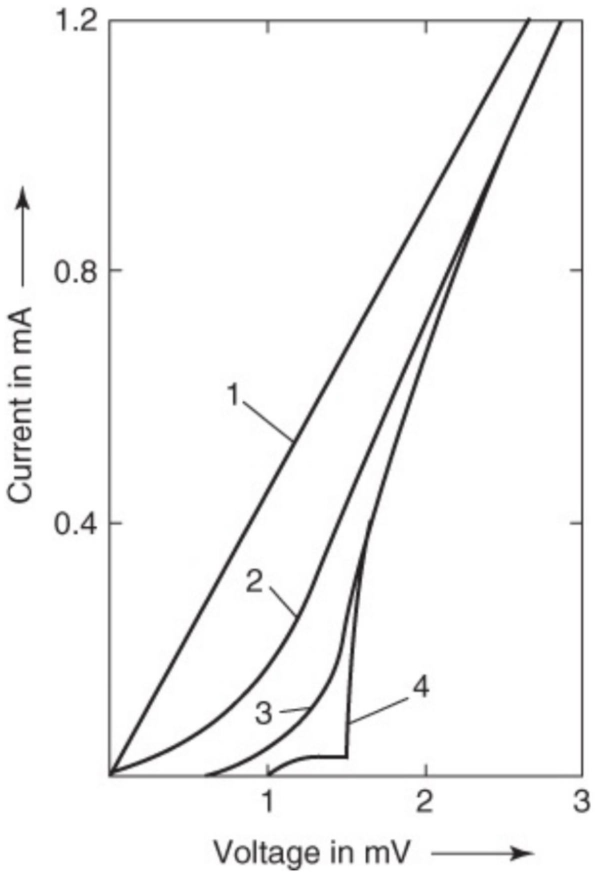
(b) Draw curves for $|u_k|^2$ and $|v_k|^2$ versus momentum $\hbar k$. **[5]**

(c) The figure below shows specific heat of tin (type-I) measured as a function of temperature.



Empty circles are the data obtained in zero field while filled circles are the data obtained in an applied field larger than the critical field. Dashed line represents the contribution expected for the electrons in the normal state and dashed-dotted line represents the lattice contribution. Estimate how large the C_e jump for a superconducting transition is relative to the C_e expected for the normal state (Express as a percentage, %). **[5]**

(d) The figure below shows voltage-current characteristic of an Al$-$Al$_2$O$_3-$Pb tunnel junction measured at different temperatures.



Curve 1 measured at 10 K, curve 2 at 4.2 K, curve 3 at 1.6 K and curve 4 at 1.0 K. Noting that Pb has a higher T_c than Al, use the figure to estimate Δ at 1.0 K and T_c for both Al and Pb, and briefly explain how it has been estimated. Then deduce the relationship between Δ at zero temperature and T_c and briefly justify your answer. **[12]**

Formula Sheet

The field due to the applied and demagnetising field for an ellipsoid
$H_i = H_a - nM$

The dimensionless parameter to distinguish type-I and II
$\kappa = \lambda/\xi$

Critical fields
$B_{c1} = B_c/\sqrt{2}\kappa;$
$B_{c2} = \sqrt{2}\kappa B_c$

Curl of the magnetic flux

$$\nabla \times \mathbf{B} = \frac{1}{r}\begin{vmatrix} \hat{r} & r\hat{\phi} & \hat{z} \\ \frac{\partial}{\partial r} & \frac{\partial}{\partial \phi} & \frac{\partial}{\partial z} \\ B_r & rB_\phi & B_z, \end{vmatrix}$$

Physical Constants and Units

Acceleration due to gravity	g	$9.81\,\mathrm{m\,s^{-2}}$
Gravitational constant	G	$6.674 \times 10^{-11}\,\mathrm{N\,m^2\,kg^{-2}}$
Ice point	T_{ice}	$273.15\,\mathrm{K}$
Avogadro constant	N_A	$6.022 \times 10^{23}\,\mathrm{mol^{-1}}$
		[*N.B.* 1 mole $\equiv 1$ *gram-molecule*]
Gas constant	R	$8.314\,\mathrm{J\,K^{-1}\,mol^{-1}}$
Boltzmann constant	k, k_B	$1.381 \times 10^{-23}\,\mathrm{J\,K^{-1}} \equiv 8.62 \times 10^{-5}\,\mathrm{eV\,K^{-1}}$
Stefan constant	σ	$5.670 \times 10^{-8}\,\mathrm{W\,m^{-2}\,K^{-4}}$
Rydberg constant	R_∞	$1.097 \times 10^{7}\,\mathrm{m^{-1}}$
	$R_\infty hc$	$13.606\,\mathrm{eV}$
Planck constant	h	$6.626 \times 10^{-34}\,\mathrm{J\,s} \equiv 4.136 \times 10^{-15}\,\mathrm{eV\,s}$
$h/2\pi$	$\hbar$	$1.055 \times 10^{-34}\,\mathrm{J\,s} \equiv 6.582 \times 10^{-16}\,\mathrm{eV\,s}$
Speed of light *in vacuo*	c	$2.998 \times 10^{8}\,\mathrm{m\,s^{-1}}$
	$\hbar c$	$197.3\,\mathrm{MeV\,fm}$
Charge of proton	e	$1.602 \times 10^{-19}\,\mathrm{C}$
Mass of electron	m_e	$9.109 \times 10^{-31}\,\mathrm{kg}$
Rest energy of electron		$0.511\,\mathrm{MeV}$
Mass of proton	m_p	$1.673 \times 10^{-27}\,\mathrm{kg}$
Rest energy of proton		$938.3\,\mathrm{MeV}$
One atomic mass unit	u	$1.66 \times 10^{-27}\,\mathrm{kg}$
Atomic mass unit energy equivalent		$931.5\,\mathrm{MeV}$
Electric constant	ϵ_0	$8.854 \times 10^{-12}\,\mathrm{F\,m^{-1}}$
Magnetic constant	μ_0	$4\pi \times 10^{-7}\,\mathrm{H\,m^{-1}}$
Bohr magneton	μ_B	$9.274 \times 10^{-24}\,\mathrm{A\,m^2}$ ($\mathrm{J\,T^{-1}}$)
Nuclear magneton	μ_N	$5.051 \times 10^{-27}\,\mathrm{A\,m^2}$ ($\mathrm{J\,T^{-1}}$)
Fine-structure constant	$\alpha = e^2/4\pi\epsilon_0\hbar c$	$7.297 \times 10^{-3} = 1/137.0$
Compton wavelength of electron	$\lambda_c = h/m_e c$	$2.426 \times 10^{-12}\,\mathrm{m}$
Bohr radius	a_0	$5.2918 \times 10^{-11}\,\mathrm{m}$
angstrom	Å	$10^{-10}\,\mathrm{m}$
barn	b	$10^{-28}\,\mathrm{m^2}$
torr (mm Hg at 0 °C)	torr	$133.32\,\mathrm{Pa}$ ($\mathrm{N\,m^{-2}}$)

Do not complete the attendance slip, fill in the front of the answer book or turn over the question paper until you are told to do so

Important Reminders

- Coats/outwear should be placed in the designated area.
- Unauthorised materials (e.g. notes or Tippex) must be placed in the designated area.
- Check that you do not have any unauthorised materials with you (e.g. in your pockets, pencil case).
- Mobile phones and smart watches **must** be switched off and placed in the designated area or under your desk. They must not be left on your person or in your pockets.
- You are not permitted to use a mobile phone as a clock. If you have difficulty seeing a clock, please alert an Invigilator.
- You are not permitted to have writing on your hand, arm or other body part.
- Check that you do not have writing on your hand, arm or other body part – if you do, you must inform an Invigilator immediately
- Alert an Invigilator immediately if you find any unauthorised item upon you during the examination.

Any students found with non-permitted items upon their person during the examination, or who fail to comply with Examination rules may be subject to Student Conduct procedures.